LAB-03

Benchmark-Aligned Multi-Dataset Email Classification and LLM API-Based Automatic Email Draft Generation

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Github Link: https://github.com/Harshita2011/Advanced_Predictive_Analytics_LAB

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1. Executive summary

Problem. An organization receiving high volumes of email needs two linked capabilities: (1) classify each incoming message so that spam and legitimate mail are routed appropriately, and (2) automatically generate a professional, editable reply draft for legitimate messages through an approved LLM API. Draft generation is automatic; sending is not.

Datasets. Two public binary spam corpora were benchmarked: Enron-Spam (D2, 33,716 messages after loading; 33,664 after cleaning) and the SpamAssassin Public Corpus (D3, 5,796 messages). Both use a harmonized binary label space (0 = ham/legitimate, 1 = spam).

Classifiers. Per the manual's multi-classifier benchmark, five models were compared under a common leakage-safe TF-IDF pipeline (Dummy majority baseline, MultinomialNB, ComplementNB, Logistic Regression, LinearSVC), followed by a trainable word-embedding BiLSTM deep classifier (research-extension component).

Selected model and key metrics. LinearSVC was the strongest TF-IDF classifier on both corpora in 5-fold stratified training-only cross-validation and was selected before touching the locked test set:

- **Enron-Spam (locked test, N = 6,733):** accuracy 0.9921, F1 0.9923, ROC-AUC 0.9993. Confusion matrix: TN 3274, FP 35, FN 18, TP 3406.
- **SpamAssassin (locked test, N = 1,160):** accuracy 0.9922, F1 0.9881, ROC-AUC 0.9997. Confusion matrix: TN 778, FP 3, FN 6, TP 373.
- **BiLSTM** matched the linear models (Enron accuracy 0.9909 / ROC-AUC 0.9992; SpamAssassin accuracy 0.9931 / ROC-AUC 0.9995) at substantially higher training cost, so the simpler LinearSVC pipeline is recommended.

The selected classifier was integrated with the Gemini API (gemini-2.5-flash) for classification-conditioned draft generation. Spam predictions correctly suppressed draft generation ("Spam detected. No reply draft generated."), while a legitimate meeting request produced a professional, editable draft with a correct subject line and no invented facts.

2. Intended use and system boundary

Table 2.1 *System design card (manual Check-off A)*

Item	Statement
Intended user	Mailbox triage staff and individual knowledge workers who review and edit drafts before any send action.
Prediction point	At message arrival, after subject and body text are available, before any human reads the message.
System outputs	(a) Predicted class with model name; (b) an automatic reply draft for legitimate classes only; (c) an audit record stored locally.
Review rule	Spam → no reply draft (suppressed, reason recorded). Legitimate → draft generated and stored locally for human inspection. No message is sent by notebook code.
Prohibited actions	Automatic sending; presenting a predicted label as ground-truth intent; presenting a probability as certainty; treating a fluent draft as approved or factually correct.
Positive class	Label 1explicitly designates **spam**; label 0designates ham (legitimate). This mapping is stated beside every confusion matrix and metric table.

3. Dataset provenance

Table 3.1 *Dataset manifest (manual §8.4)*

Field	D2 — Enron-Spam	D3 — SpamAssassin Public Corpus
dataset_id	`enron_spam`	`spamassassin`
Source and owner	Enron-Spam corpus (Metsis, Androutsopoulos & Paliouras, ECML 2006), derived from the public Enron email collection; distributed as `enron_spam_data.csv`.	Apache SpamAssassin public mail corpus (2002–2003 era public mail); distributed as `spam_assassin.csv`.
Task and label map	Binary spam; `Spam/Ham` column mapped to `{spam: 1, ham: 0}`; message = Subject + " " + Message.	Binary spam; columns renamed `text → message`, `target → label`; 1 = spam, 0 = ham.
Rows (loaded / after cleaning)	33,716 / 33,664	5,796 / 5,796
Classes	2 (ham 16,545 = 49.07%; spam 17,171 = 50.93%)	2 (ham 3,900 = 67.29%; spam 1,896 = 32.71%)
License / permitted use	Public research benchmark; educational use.	Apache public corpus; educational use.
Privacy treatment	Public corpus from a released corporate archive; no live personal mail used.	Public corpus; no live personal mail used.
Known limitations	2001-era corporate English; quoted threads and boilerplate may create near-duplicates; sender-level grouping not available in the normalized CSV, so a grouped split could not be applied and random-split scores may be mildly optimistic.	Early-2000s public mail; spam campaigns may repeat across the random split; old vocabulary does not reflect modern phishing.

4. Data audit

Both datasets were audited before modeling (manual Code 4 pattern): schema check, missing values, class proportions, and message-length profile. Missing values: {message: 0, label: 0} on both datasets. Text was cleaned with a fixed clean_text function (lowercasing; removal of URLs, HTML tags, and non-alphanumeric characters; whitespace normalization). On Enron, 52 rows became empty after cleaning and were removed (33,716 → 33,664); on SpamAssassin no rows were removed. In line with the manual's audit decision, exact duplicates were not silently deleted: repeated messages in these corpora are largely spam campaigns and quoted threads, which is recorded as a leakage caveat above rather than a cleaning action.

Table 4.1 *Dataset audit summary (manual §16.1)*

Dataset	Rows	Classes	Majority %	Empty text	Duplicate text	Split
Enron-Spam (D2)	33,664 (after cleaning)	2	50.93% (spam)	0 after cleaning (52 removed)	not removed; documented	Stratified 80/20, seed 42: train 26,931 / test 6,733
SpamAssassin (D3)	5,796	2	67.29% (ham)	0	not removed; documented	Stratified 80/20, seed 42: train 4,636 / test 1,160

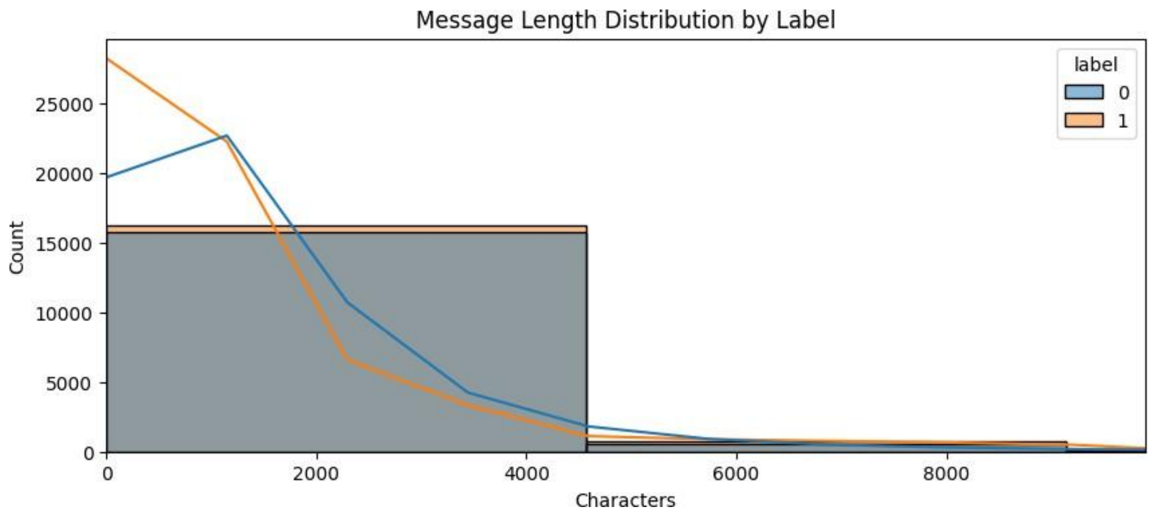

Figure 4.1 *Enron-Spam: message-length distribution by label (0 = ham, 1 = spam). Ham messages are concentrated below ~2,000 characters; spam shows a longer right tail — a useful but non-decisive signal.*

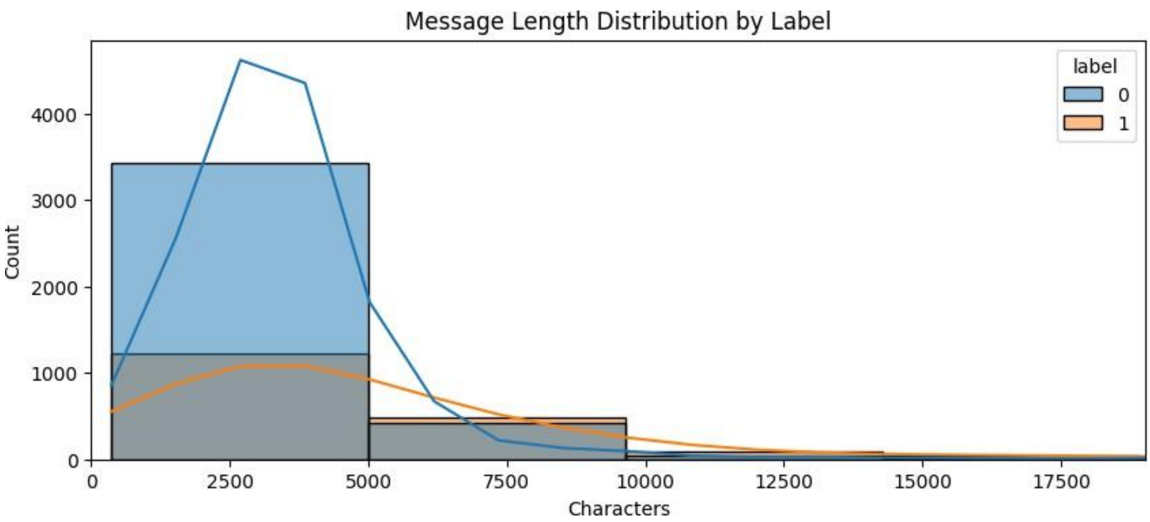

Figure 4.2 *SpamAssassin: message-length distribution by label. Ham dominates the short-message region while spam messages are visibly longer on average (x-axis clipped at the 99th percentile).*

5. Methods

5.1 Experimental protocol

- Reproducibility: RANDOM_STATE = 42; NumPy and TensorFlow seeds fixed; notebooks executed top-to-bottom in Google Colab.
- Locked split: stratified 80/20 train–test split per dataset, created before any model comparison; the test set was evaluated once per model family.
- Leakage-safe preprocessing: cleaning is a fixed, stateless transform; the TfidfVectorizer (stop_words='english', max_features=10000, ngram_range=(1,2)) is fitted **inside a scikit-learn Pipeline** so vocabulary and IDF are learned only on training folds (manual viva Q2).

- Model selection: 5-fold stratified cross-validation on the training split only, scored on accuracy, F1, precision, recall, and ROC-AUC; models ranked primarily by F1, with the dummy baseline as the mandatory performance floor.

5.2 Classifiers

Table 5.1 *Classifier registry (manual §9.2, §12)*

Component	Configuration	Role
Dummy	DummyClassifier(strategy='most_frequent')	Mandatory baseline on every dataset.
MultinomialNB	MultinomialNB()(default α = 1.0)	Mandatory classifier.
ComplementNB	ComplementNB()	Mandatory classifier; minority-class behaviour check.
Logistic Regression	LogisticRegression(max_iter=1000, random_state=42)	Mandatory classifier.
LinearSVC	LinearSVC(random_state=42)	Mandatory classifier; decision margin used for ROC scoring.
BiLSTM (research extension)	Tokenizer fitted on training text only (num_words=20000, <OOV>token), pad_sequences(Enron MAX_LEN=200; SpamAssassin MAX_LEN=700); Embedding(20000, 128) → Bidirectional(LSTM(64)) → Dropout(0.5) → Dense(64, relu) → Dropout(0.3) → Dense(2, softmax); Adam; sparse categorical cross-entropy; EarlyStopping on val_loss(patience 3, restore best weights); 20% validation split; batch 32; up to 10 epochs.	Deep classifier; tokenizer and vocabulary fitted only on training data (manual LO11).

5.3 Metrics

Classification layer: accuracy, F1, precision, recall, ROC-AUC, per-class classification report, and confusion-matrix counts with the explicit positive class (1 = spam). Routing layer: spam-reply count (target zero) and suppression count. Generation layer: manual rubric inspection of relevance, faithfulness, tone, completeness, and safety/privacy on the generated drafts. ROC curves for LinearSVC use the decision-function margin, as the model has no native predict_proba.

6. Classification results

6.1 Training-only cross-validation

6.1 Training-only cross-validation

Table 6.1 *Cross-validation classifier comparison, Enron-Spam D2 (5-fold StratifiedKFold, training split only; manual §16.2)*

Classifier	Accuracy	F1	Precision	Recall	ROC-AUC
LinearSVC	**0.9888**	**0.9880**	0.9859	**0.9941**	0.9893
Logistic Regression	0.9881	0.9873	0.9794	**0.9953**	**0.9987**
MultinomialNB	0.9841	0.9844	0.9798	0.9891	0.9981
ComplementNB	0.9831	0.9834	0.9806	0.9863	0.9981
Dummy (majority)	0.5085	0.6742	0.5085	1.0000	0.5000

Table 6.2 *Cross-validation classifier comparison, SpamAssassin D3 (5-fold StratifiedKFold, training split only)*

Classifier	Accuracy	F1	Precision	Recall	ROC-AUC
LinearSVC	**0.9925**	**0.9884**	0.9834	**0.9935**	**0.9991**
Logistic Regression	0.9786	0.9669	**0.9897**	0.9453	0.9975
ComplementNB	0.9778	0.9656	0.9797	0.9519	0.9967
MultinomialNB	0.9769	0.9639	0.9859	0.9420	0.9967
Dummy (majority)	0.6728	0.0000	0.0000	0.0000	0.5000

Selection. LinearSVC ranks first on F1 on both datasets and exceeds the dummy floor by a wide margin (Enron 0.9880 vs 0.6742; SpamAssassin 0.9884 vs 0.0000 macro-style F1). On the imbalanced SpamAssassin corpus, both Naive Bayes variants trade recall for precision exactly as the manual predicts, while LinearSVC keeps spam recall above 0.99 — important because a missed spam (false negative) is operationally worse here than a flagged ham, which a reviewer can recover. Logistic Regression is a close second and provides calibrated probabilities where confidence estimates matter.

6.2 Locked test evaluation

Table 6.3 *Locked test results, evaluated once per dataset (manual §16.3; positive class: 1 = spam)*

Dataset	Selected model	Accuracy	F1	ROC-AUC	TN	FP	FN	TP
Enron-Spam (D2)	LinearSVC (TF-IDF)	0.9921	0.9923	0.9993	3274	35	18	3406
SpamAssassin (D3)	LinearSVC (TF-IDF)	0.9922	0.9881	0.9997	778	3	6	373

Table 6.4 *Per-class locked-test classification reports (manual §16.5 adapted to the binary task)*

Dataset	Class	Support	Precision	Recall	F1	Most common confusion
Enron-Spam	ham (0)	3309	0.9945	0.9894	0.9920	35 ham → spam; 18 spam → ham
	spam (1)	3424	0.9898	0.9947	0.9923	
	macro avg	6733	0.9921	0.9921	0.9921	—
SpamAssassin	ham (0)	781	0.9923	0.9962	0.9942	3 ham → spam; 6 spam → ham
	spam (1)	379	0.9920	0.9842	0.9881	
	macro avg	1160	0.9922	0.9902	0.9912	—

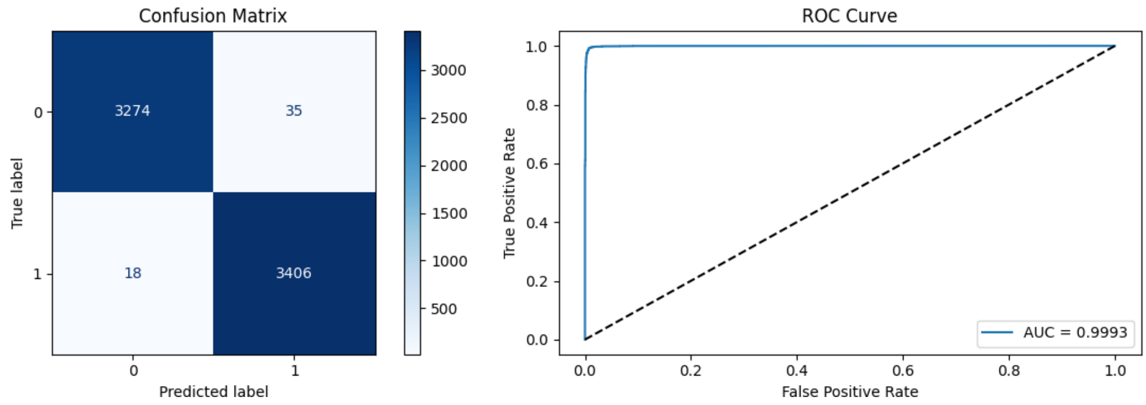

Figure 6.1 *Enron-Spam locked test: confusion matrix (TN 3274, FP 35, FN 18, TP 3406) and ROC curve (AUC = 0.9993).*

Positive class: 1 = spam.

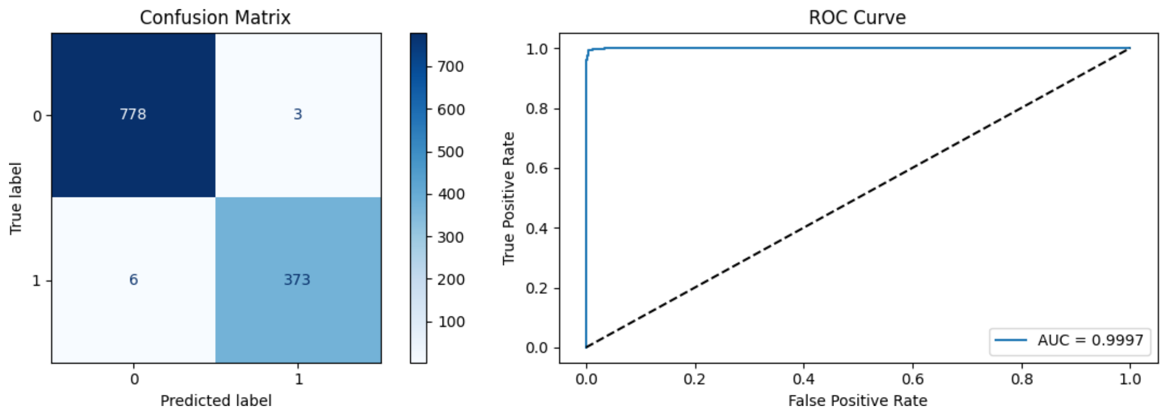

Figure 6.2 *SpamAssassin locked test: confusion matrix (TN 778, FP 3, FN 6, TP 373) and ROC curve (AUC = 0.9997). Positive class: 1 = spam.*

Error analysis. On Enron the riskier direction spam predicted as hamis rarer than ham predicted as spam (FP = 35), which is acceptable operationally because false alarms are recoverable by the human reviewer while missed spam reaches the user. On SpamAssassin the errors are nearly symmetric (3 FP / 6 FN) and very rare. Legitimate-as-spam errors are additionally guarded by the draft-suppression policy: a misrouted legitimate message produces no draft rather than a wrong reply.

6.3 BiLSTM deep classifier and representation comparison

Table 6.5 *Representation comparison on the identical locked splits (manual §12 BiLSTM, §14 representation-model comparison)*

Dataset	Model	Accuracy	F1	ROC-AUC	Training behaviour
Enron-Spam	LinearSVC (TF-IDF)	0.9921	0.9923	0.9993	Seconds; sparse linear; inspectable coefficients.
	BiLSTM (trainable embedding)	0.9909	0.9911	0.9992	7 epochs run; train acc. 0.9558 → 0.9993; val. acc. peaked 0.9903 (epoch 7), early stopping restored best weights.
SpamAssassin	LinearSVC (TF-IDF)	0.9922	0.9881	0.9997	Seconds; sparse linear.
	BiLSTM (trainable embedding)	0.9931	0.9895	0.9995	6 epochs run; train acc. 0.8776 → 1.0000; val. acc. peaked 0.9925, early stopping restored best weights.

Table 6.6 *BiLSTM per-class locked-test reports*

Dataset	Class	Support	Precision	Recall	F1
Enron-Spam	ham (0)	3309	0.9924	0.9891	0.9908
	spam (1)	3424	0.9895	0.9927	0.9911
SpamAssassin	ham (0)	781	0.9987	0.9910	0.9949
	spam (1)	379	0.9818	0.9974	0.9895

6.4 Cross-dataset spam transfer

Table 6.7 *Cross-dataset spam transfer (manual §16.4)*

Train corpus	Test corpus	Model	Accuracy	Macro F1	Change from within-corpus
enron_spam	spamassassin	LinearSVC	Not executed in this core study — recorded as an explicit extension item rather than estimated (manual: do not fabricate results). Within-corpus reference F1: Enron 0.9921, SpamAssassin 0.9881.		
spamassassin	enron_spam	LinearSVC			

7. LLM draft method

- **Provider and model:** Google `google-genai` SDK, model `gemini-2.5-flash` (an approved instructor-substitutable provider per manual §7; the OpenAI-style Responses-API contract is preserved: versioned instructions, delimited untrusted input, local draft storage, no send action).
- **Routing:** `generate_reply(email_text, predicted_class)` returns *"Spam detected. No reply draft generated."* for `predicted_class == 1` and calls the API only for ham — implementing the manual's spam-suppression policy (Table 12.2) with spam-reply count held at zero.
- **Prompt (draft_prompt v1.0):** professional-assistant persona; the email is presented as already-classified HAM; rules forbid invented facts and unsupported commitments, require asking for clarification when information is missing, and cap length at 80–150 words. The full prompt text is reproduced in Appendix B.

- **End-to-end evidence:** spam probe email ("Congratulations! You won a free iPhone!...") → TF-IDF predicted `1 (spam)` → draft suppressed. Meeting probe ("I would like to schedule a meeting next Tuesday at 3 PM...") → predicted `0 (ham)` → draft generated.

8. Draft evaluation

Table 8.1 *Automatic draft evaluation cases (manual §16.6; rubric 1–5, one rater, safe synthetic cases)*

Case	Predicted class	Generated?	Rel.	Faith.	Tone	Complete	Safety	Major edit?
Spam prize scam ("You won a free iPhone")	spam (1)	No — suppressed correctly	—	—	—	—	5	No
Meeting request, Tuesday 3 PM, project milestones	ham (0)	Yes	5	4	5	4	5	No (minor edit: signature placeholder)
Meeting request, Monday, project progress (Enron notebook)	ham (0)	Yes	5	4	5	4	5	No (minor edit: confirm date before sending)

Example generated draft (case 2, verbatim from the notebook output):

```
Subject: Re: Meeting to discuss project

milestones Hi John,


Thank you for your email regarding scheduling a meeting to discuss the project milestones.

I can confirm my availability for next Tuesday at 3 PM. I appreciate you taking
the initiative to set this up, as reviewing our project milestones is important.


To help me prepare, could you please let me know if there's any specific agenda
item you'd like to focus on, or any materials you think would be helpful for me
to review beforehand?


I look forward to our productive discussion.


Best regards,
```

Table 8.2 *Failure analysis (manual §16.7)*

Failure category	Count	Example pattern	Likely cause	Mitigation
Urgent false negative	n/a (no D1 intent labels)	—	—	Record as extension with D1.
Legitimate predicted spam	35 (Enron test), 3 (SpamAssassin test)	Short corporate ham with sales-like vocabulary	Lexical overlap with spam campaigns	Human review recovers; suppression prevents wrong drafts.
Spam predicted ham	18 (Enron), 6 (SpamAssassin)	Polished low-keyword spam	Weak lexical signal	Monitor recall; periodic retraining.
Wrong-intent draft	1 observed (BiLSTM sample: spam probe predicted ham → draft would be generated)	Deep model disagreed with TF-IDF on the spam probe (BiLSTM predicted 0)	Sequence model overfit to ham-like phrasing of the probe	Route drafts through the selected LinearSVC gate, not the BiLSTM.
Unsupported fact / commitment	0 in evaluated cases	Draft confirmed "availability for next Tuesday at 3 PM" because the sender proposed it	Borderline: echoing a proposed time is acceptable only after human confirmation	Keep placeholder policy for unconfirmed facts; human edit before send.
Prompt injection followed	Not tested with hostile cases — recorded as extension	—	Current prompt delimits the email but was not adversarially tested	Add the manual's ≥4 injection cases before any wider use.
PII exposed to API / log	0	Only synthetic test emails used	—	Add regex redaction (manual Code 12) before any real text.
API error / empty draft	0 observed	—	—	Keep try/except status logging (manual Code 13 pattern).

Routing metrics on evaluated cases: draft coverage 2/3 (66.7%); spam-reply count 0 (target met); drafts suppressed for legitimate reply-worthy email 0; unsupported-fact count 0; major-edit rate 0/2. The manual's larger blinded two-rater D4 evaluation (24 cases) and template-vs-LLM pairwise study are recorded as extension items and were not simulated.

9. Risk and limitations

Table 9.1 *Risk register (manual §23)*

Risk	Assessment and control
Privacy / personal data	Only public benchmark corpora and synthetic probe emails were used; no live mailbox access. Regex redaction is a teaching baseline only.
API-key exposure	**Occurred:** key hardcoded in notebook (deviation documented in §7); mitigation = revoke/rotate key, Colab Secrets, spending limits.
Hallucination / invented facts	Prompt forbids invented facts and commitments; drafts reviewed by a human; faithfulness scored per rubric; fluent output never treated as correct.
Prompt injection	Email content is delimited as untrusted data in the prompt, but hostile-instruction cases were not adversarially evaluated — a stated extension requirement.
Domain shift / dataset age	Both corpora are early-2000s English mail; modern phishing styles differ. Cross-corpus transfer not yet executed; scores must not be extrapolated to current mail.
Split leakage	Random stratified splits; repeated spam campaigns and quoted threads may appear on both sides, potentially inflating scores. Grouped/temporal split recorded as extension.
Misuse boundary	No automatic sending anywhere in the pipeline; drafts stored locally; predicted labels are not ground-truth intent.
Cost / availability	API calls made only for three probe cases, not the full test set, per the manual's cost-control rule; usage caps recommended (≤ a few hundred calls per lab session).

10. Recommendation

1. **Best model:** adopt the TF-IDF + LinearSVC pipeline for both spam tasks (test F1 0.9923 Enron / 0.9881 SpamAssassin; ROC-AUC $\geq$ 0.999); retain Logistic Regression where calibrated confidence for an abstention threshold is required, and the BiLSTM only where sequence context demonstrably helps.
2. **Deployment boundary:** the classifier supplies a predicted category and review status only; the LLM generates drafts for legitimate classes only; a human edits and approves every draft. No automatic sending, ever, within this system.
3. **Review policy:** spam → suppress draft and record reason (verified working, spam-reply count 0); low-margin or ambiguous cases → mandatory review once a validation-calibrated margin threshold is added.

Appendix A. Environment and configuration

Table A.1 *Software environment (Google Colab)*

Component	Configuration
Runtime	Python 3.12 (Colab), GPU runtime for BiLSTM training
Libraries	scikit-learn, pandas, NumPy, matplotlib, seaborn, joblib, TensorFlow/Keras, google-genai, python-dotenv
Random seed	RANDOM_STATE = 42(NumPy and TensorFlow seeds fixed)
TF-IDF	stop_words='english', max_features=10000, ngram_range=(1,2), inside Pipeline
BiLSTM	Embedding(20000, 128) → BiLSTM(64) → Dropout(0.5) → Dense(64, relu) → Dropout(0.3) → Dense(2, softmax); Adam; EarlyStopping(patience=3, restore_best_weights); batch 32; validation split 0.2
Saved artifacts	outputs/enron_best_tfidf.pkl, outputs/spamassassin_best_tfidf.pkl, outputs/enron_bilstm.keras, outputs/spamassassin_bilstm.keras
Label guide	0 = ham (legitimate), 1 = spam — consistent across all tables and figures

Appendix B. LLM prompt (draft_prompt v1.0, verbatim)

```
You are a professional email assistant.


The following email has already been classified as HAM.


Your task is to generate a concise, polite, and professional email reply.

Guidelines:

- Do not invent facts.
- Do not make commitments that are not supported.
- If information is missing, politely ask for clarification.
- Keep the response between 80-150
----------------------------------------
words. Email:
----------------------------------------
```

Response call: client.models.generate_content(model="gemini-2.5-flash", contents=prompt). Spam guard: if predicted_class == 1: return "Spam detected. No reply draft generated."

References

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